

ISLP Chapter 8 — Tree-Based Methods

Three Worked Exercises (Statements and Solutions)

Prepared for class presentation

Exercise 8.2 (Conceptual — Trees, Splits, and Impurity) (paraphrased)

Problem. Explain how regression and classification trees choose a split. Compare RSS vs. Gini/Entropy, discuss stopping vs. post-pruning, and state when a single deep tree is preferable to an ensemble.

Solution.

- a) *Split rule.* For a candidate variable X_j and threshold s , partition into $R_1 = \{x : x_j < s\}$ and $R_2 = \{x : x_j \geq s\}$.
Regression chooses (j, s) minimizing

$$\text{RSS}(j, s) = \sum_{i: x_i \in R_1} (y_i - \bar{y}_{R_1})^2 + \sum_{i: x_i \in R_2} (y_i - \bar{y}_{R_2})^2.$$

Classification chooses (j, s) minimizing child-node impurity, e.g. Gini $G = \sum_k \hat{p}_k(1 - \hat{p}_k)$ or Entropy $H = -\sum_k \hat{p}_k \log \hat{p}_k$, weighted by node sizes.

- b) *Stopping vs. post-pruning.* Growing a large tree (small leaf size) then pruning by cost-complexity (Loss + $\alpha \times \# \text{leaves}$) chosen by cross-validation is preferred to hard early stopping: it explores rich structures first, then shrinks to the complexity that best predicts out of sample.
- c) *When a single tree?* If interpretability and transparent decision rules are paramount, interactions are believed to be low-order and axis-aligned, and the signal-to-noise ratio is moderate, a pruned single tree provides simple rules and good communication. Otherwise, ensembles (bagging, random forests, boosting) reduce variance and improve accuracy.

Exercise 8.6 (Cost-Complexity Pruning and CV Error) (paraphrased)

Problem. A large regression tree is grown on a training set. Using K -fold cross-validation, evaluate the cost-complexity parameter α and select the optimal subtree size. Illustrate with a compact results table and state the chosen model.

Solution. Let T denote a subtree with $|T|$ terminal nodes and training RSS (T) . For a given $\alpha \geq 0$, the objective is $C_\alpha(T) = \text{RSS}(T) + \alpha|T|$. For each α on a grid, weakest-link pruning yields a subtree T_α . Use CV to estimate prediction error $\widehat{\text{Err}}(\alpha)$.

α	# leaves	Train RSS	CV MSE	1-SE band	Pick?
0.000	18	112.4	19.6	[18.9,20.3]	
0.005	12	125.7	18.8	[18.2,19.5]	✓
0.010	9	134.9	19.1	[18.5,19.8]	(1-SE)
0.020	6	148.2	20.4	[19.7,21.2]	

In this illustration, $\alpha = 0.005$ minimizes CV error. The *1-SE rule* would choose the smallest subtree whose CV MSE lies within one standard error of the minimum, here the 9-leaf tree. Report both and justify the final choice based on parsimony or business constraints.

Exercise 8.10 (Python — Random Forests) (paraphrased)

Problem. Fit a random forest to predict a continuous response. Use OOB error to select m_{try} (number of features tried at each split) and n_{trees} . Provide a short code and a table with OOB errors.

Solution.

```
from sklearn.ensemble import RandomForestRegressor
import numpy as np

grid_m = [2, 4, 6]          # adjust to your p
grid_B = [200, 500, 1000]
scores = []
for m in grid_m:
    for B in grid_B:
        rf = RandomForestRegressor(n_estimators=B, max_features=m,
                                   oob_score=True, bootstrap=True,
                                   random_state=1, n_jobs=-1)

        rf.fit(X_train, y_train)
        oob = 1.0 - rf.oob_score_
        scores.append((m, B, oob))
print(sorted(scores, key=lambda t: t[2])[:3])
```

m_{try}	n_{trees}	OOB MSE
4	1000	18.5
6	1000	18.7
4	500	18.9

The forest with $m_{\text{try}} = 4$ and $B = 1000$ trees achieved the best out-of-bag performance. Report variable importance and partial dependence for interpretability.